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TECHNICAL NOTE

Thermoplastic elastomers for microfluidics: Towards a high-throughput fabrication method of multilayered microfluidic devicesEmmanuel Roy,^{*} Jean-Christophe Galas and Teodor Veres

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Multilayer soft lithography of polydimethylsiloxane (PDMS) is a well-known method for the fabrication of complex fluidic functions. With advantages and drawbacks, this technique allows fabrication of valves, pumps and micro-mixers. However, the process is inadequate for industrial applications. Here, we report a rapid prototyping technique for the fabrication of multilayer microfluidic devices, using a different and promising class of polymers. Using styrenic thermoplastic elastomers (TPE), we demonstrate a rapid technique for the fabrication and assembly of pneumatically driven valves in a multilayer microfluidic device made completely from thermoplastics. This material solution is transparent, biocompatible and as flexible as PDMS, and has high throughput thermoforming processing characteristics. We established a proof of principle for valving and mixing with three different grades of TPE using an SU-8 master mold. Specific viscoelastic properties of each grade allow us to report enhanced bonding capabilities from room temperature bonding to free pressure thermally assisted bonding. In terms of microfabrication, beyond classically embossing means, we demonstrate a high-throughput thermoforming method, where TPE molding experiments have been carried out without applied pressure and vacuum assistance within an overall cycle time of 180 s. The quality of the obtained thermoplastic systems show robust behavior and an opening/closing frequency of 5 Hz.

1. Introduction

Today, microfluidic devices are made from various materials but are mostly made of silicon, glass or polymers. Among these materials, PDMS offers an easy solution for prototyping. This elastomer was introduced by Whitesides and is well-known in the multilayer configuration developed by Quake for the integration of micro-valves, and is widely used in laboratories.^{1,2} However, such systems have mostly been shown to function in conceptual ways, yet their commercialisation and widespread use have been hindered by a number of challenges including material costs, low-cost fabrication and bonding means, and assembling technologies. Until now, the catalogue of existing polymers has not been extensively explored for multilayered microfluidic device fabrication and applications.³ Several attempts to use new materials that could replace PDMS have been undertaken over the last few years. For example, Rolland *et al.*⁴ demonstrated the possibility of structuring microchannels along with the integration of valving capabilities using soft, photocurable perfluoropolyethers. Fiorini *et al.*⁵ introduced a thermoset polyester resin which allows the creation of solvent resistant microfluidic devices within 3 h using a process that involves casting, UV

exposure and thermal curing steps. In this work we focused on TPE, a class of material in which elastomeric properties of vulcanized rubber (*e.g.*, PDMS) are embodied with the ease of processing of thermoplastic materials such as poly(methyl-methacrylate) (PMMA), polycyclic olefin (PCO) and polycarbonate (PC), among others.⁶ We have recently identified commercially available thermoplastic elastomers and illustrated some of their specific advantages and benefits for microfluidic devices.^{7–10} We have succeeded in handling DNA and protein solutions for microfluidic spotting and we performed human cell cultures onto isothermally microstructured TPE surfaces.^{7,8} A description of TPE materials has been published recently and allows a comprehensive understanding of TPE materials according to rheological molding specificities and detailed bonding properties.⁹ We have also explored TPE material solutions for high temperature applications, supporting, for example, polymerase chain reaction-based amplification of DNA.¹⁰ Table 1 lists the essential characteristics of PDMS, TPE and hard TP (thermoplastic) materials. Given these properties, we believe that thermoplastic elastomers satisfy most of the requirements needed for the development of an industrial process for low-cost, large scale fabrication of microfluidic devices. The combination of elastomeric and thermoplastic properties makes these materials potential substitutes to PDMS and/or hard TP polymers that are commonly employed. Unlike PDMS, TPE can be used in the form of extruded sheets that provide off-the-shelf availability

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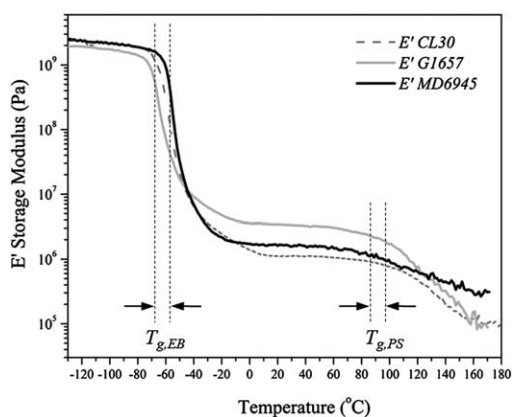
Table 1 (a) Table comparing properties of PDMS, TPE and hard thermoplastics (TP) materials properties for microfluidic technology

	PDMS	TPE	TP
Optical transparency	++	+	+
Young modulus	0.1–5 MPa	1–100 MPa	1–10 GPa
Thermoforming cycle	very slow	fast	fast
Gas permeability	high	moderate	low
Cost (\$/Lbs)	10–50	0.1–5	0.1–5
Reversible bonding	+	+	impossible
Irreversible bonding	+	++	—
Resolution	50 nm	10 nm(?)–100 nm	10 nm
Ease of de-molding	++	++	—
References	(3,11,12)	(6,8,9)	(3,13–15)

without the necessity of performing any pre-compounding step. To the best of our knowledge, TPE has never been used for multilayer valving and mixing in microfluidic devices. In that direction, and pursuing our approach and strategy to demonstrate the advantages and benefits of thermoplastic elastomer materials for microfluidics, in this paper we extend this work by first presenting a novel rapid prototyping technique for the fabrication and assembly of pneumatically driven valves in a multilayer microfluidic device made completely from thermoplastics.

2. Materials and methods

Among the different classes of TPE, we focused on the block copolymers category and we selected commercial grades of (styrene)–(ethylene/butylene)–(styrene) (SEBS) block copolymers: Versaflex CL30 (GLS Corp., McHenry, IL, USA), MD6945 and G1657 (Kraton Polymers, Houston, TX, USA). The pellets were extruded to form films of 0.2–3.0 mm in thickness and several meters long. Extruded films can be stored over long periods of time (*e.g.*, several years) without any notable degradation, making it possible to use the material on demand without any time consuming pre-compounding step. Advantages of TPE material over fabrication, assembling and bonding issues, which are of importance for microfluidic applications, lay principally on their viscoelastic and mechanical properties.^{9,16} Dynamic mechanical thermal analyses have been carried out,

**Fig. 1** Results of dynamic mechanical thermal analyses obtained on 1 mm thick polymer films (G1657, CL30 and MD6945 TPE grades).

and storage (E') moduli *versus* temperature are shown in Fig. 1. Experiments were performed on 1 mm polymer sheets with an ARES LS2 rheometer, upon heating from -120 to 180 °C. Moduli were determined at 1 Hz with a strain of 0.02% and a heating rate of 2 °C min^{-1} . As block copolymer materials, the TPEs exhibit two glass transition temperatures corresponding to the ethylene/butylene (EB) soft block ($T_{g,EB} \approx -65 \pm 7$ °C) and to the styrene rigid blocks ($T_{g,PS} \approx 90$ °C), respectively. The negative value of $T_{g,EB}$ predicts liquid like behavior of the materials. Indeed, as with PDMS, which similarly displays a low glass transition temperature (≈ -125 °C),^{17,18} the selected TPEs form a spontaneous and conformable close contact with flat substrates, generating water tight sealing. Additionally, the soft blocks provide bonding capabilities above their glass transition which implies that even at room temperature, the polymer chains can re-organize according to the contacted surface.⁹ The bonding strength is variable and dependent on time and temperature; an irreversible bonding was obtained at room temperature after 2 days when two layers of MD6945 material were joined. For the CL30 and G1657 materials, we report an irreversible bonding with a thermal treatment of 3 min at 100 °C. Above $T_{g,EB}$, E' storage reach a so-called rubber plateau; moduli of 1.1, 1.6 and 3.4 MPa at 25 °C are measured for CL30, MD6945 and G1657, respectively. These mechanical properties are in line with our objectives to select thermoplastics materials which are as soft as PDMS, and their specific block copolymer structure allow a material in the vicinity of 10–15% of styrene hard blocks content to be obtained, which are 1000 times softer than neat polystyrene. Above the $T_{g,PS}$, E' moduli reach a quasi-plateau region in the 160 – 165 °C vicinity to a steady value of 0.8 – 3.1×10^5 MPa. Therefore, an embossing temperature of 170 °C provides a good compromise between the lowest E' moduli values that facilitate replication and a processing temperature to minimize mold damage and thermal expansion. The devices presented herein, similar to those previously described by S. Quake *et al.* in 2000,² are the result of the assembly of two structured polymer layers that are easily bonded due to TPE intrinsic viscoelastic properties. Fig. 2 displays the protocol used for the fabrication and assembly of the TPE valving multilayered device. Two layers of TPE have been thermally embossed using SU-8 molds and the thickness of the thin microfluidic layer was controlled using calibrated spacers placed on the mold surface (Fig. 2a). The thick layer with embedded control channels was also structured over an SU-8 mold (Fig. 2b), easily peeled-off due to its elastomeric properties and then successively placed and aligned over the thin layer (Fig. 2c). The intersections between the upper and lower level channels create membranes that can be actuated by applying a pressure (Fig. 2d). The two TPE layers were irreversibly bonded together either with a thermal treatment in the case of CL30 and G1657 materials and at room temperature for the MD6945 material. Finally, the TPE bi-layer structure was peeled-off from the mold, access holes were punched and the stack was thermally bonded on a flat poly (cyclo)olefin polymer substrate (Fig. 2d).

3. Results and discussion

For the thin layer fabrication, with reflow of SU-8 photoresist being impossible, we developed a method to obtain a rounded

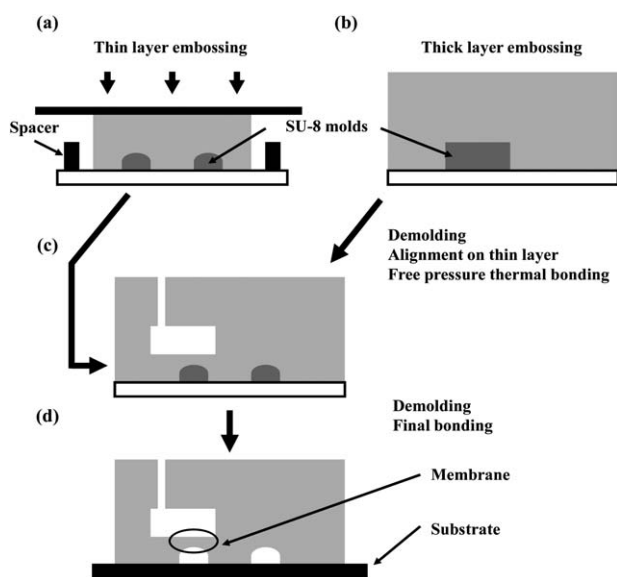


Fig. 2 Step-by-step fabrication process of the multi-layer TPE microfluidic device: (a,b) thin and thick layers embossing, (c) layers alignment and bonding, (d) device sealing.

profile (which is essential for valve operation) in order to create a thermally stable SU-8 master for molding. A PDMS layer is first structured over a rounded mold done in AZ9260 positive resist. Molded PDMS is then treated with an anti-adhesive solution and then all over poured with a Microchem SU8-5 resist. A silicon wafer is put against the resist and a low pressure is applied over the PDMS. After two soft bakes at 65 and 95 °C for 15 min each, the resist is exposed to UV light. A final post-exposure bake at 120 °C for 30 min complete the curing. Upon removal of the PDMS layer, high quality rounded profiles of SU-8 are revealed. Finally, a piece of 200 μm thick extruded TPE sheet was placed onto the mold and embossed using an anti-adhesive treated silicon wafer as a counter plate. In order to calibrate the final thickness of the layer, 60 μm spacers were used. Considering a mean central height of 35 μm for fluidic channels, it allows the creation of a 25 μm thick membrane in between control and fluidic channels.

All three TPE materials were molded under a primary vacuum with an applied pressure of 4 kN for 10 min at 170 °C. Upon removal, the membrane consistently remains on the side of the resist mold due to the resist structures which increase roughness and surface area. This is convenient for the next assembling step.

Considering the thick layer fabrication, we started from a 3 mm thick extruded TPE sheet. All three TPE materials were embossed in the same conditions as for the thin layer using an SU-8 mold with 30 μm features height. Fig. 3 a, b & c display scanning electron micrographs (SEM) of dense features of lines, circles and square arrays for G1657, MD6945 and CL30, respectively. Structures show well-defined shapes and excellent surface qualities. Additionally, classical defects encountered for hard thermoplastics^{19,20} such as edge damage and asymmetric pull-off of plastics, in part due to interfacial friction between the polymer and the mold upon de-molding, are absent. We think this is due to the capabilities of TPE materials to sustain deformation through their compression set characteristics. Based on

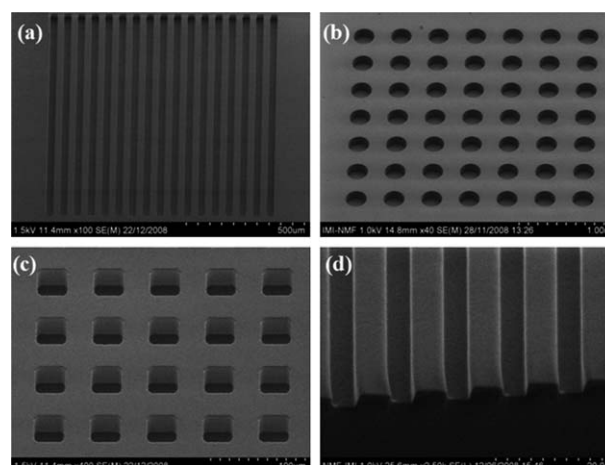


Fig. 3 SEM pictures of embossed TPE materials (4 kN at 170 °C for 10 min): (a) G1657 (b) MD6945 and (c) CL30 materials; (d) free pressure embossing of 3 mm polymer sheet of CL30 SEBS TPE material. Experimental conditions: 170 °C under its own weight on a simple laboratory hot plate equipment.

our present experience, we have been able to use an SU-8 master mold between 20 and 50 times without compromising the quality of the resultant replicas.

Moreover, the softer CL30 TPE grade offers a significant improvement in the ease of the thermoforming process. Indeed, using a common hot plate, it can be embossed in 3 min at 170 °C under its own weight. Here, a 3 mm thick CL30 sheet is placed on the mold inducing a sufficient pressure (~ 50 Pa) to ensure polymer displacement and filling of the mold cavities. Fig. 3d presents the excellent replication fidelity of an array of 5 μm wide lines. During the process, the few remaining trapped air bubbles between the mold and the polymer sheet are evacuated after 2 min of heating. We believe that in addition to its low mechanical modulus and rheology properties, the gas permeability of CL30 material is crucial to achieve this simple process. From the literature,^{21,22} it has been reported that TPE styrenic materials have gas permeability coefficients 20 to 400 times lower than PDMS.

Finally, the stack mold/CL30 was cooled down at room temperature and de-embossing was performed manually by peeling-off the polymer layer. Additional technical and scientific details related to this thermoforming process, which we named soft thermoplastic lithography, are described and discussed in terms of polymer rheology in recently published work.⁹

In terms of device assembly, the thick layer was positioned and aligned with respect to the channel network of the thin layer. At this step, the bonding strength still provides reversibility and allows for convenient manipulation to correct misalignment. As discussed above, for CL30 and G1657 material, irreversible bonding can be achieved with a 3 min heating step at 100 °C, while for MD6945, irreversibility occurs at room temperature after two days. Finally, access holes were punched and a last bonding step was done between the bi-layer structure and a poly (cyclo)olefin polymer substrate (*i.e.* 100 °C for 3 min) (Fig. 4a).

For microfluidic validation, a home-made pneumatic interface was used to apply 0–30 PSI pressure and thus drive fluids. Fig. 4b displays optical images of the opened and closed channel through

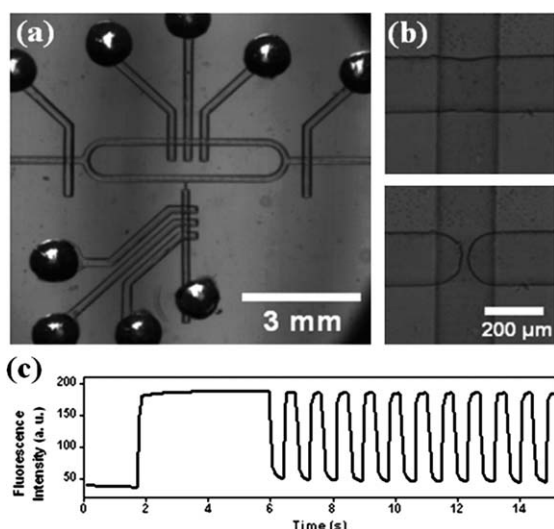


Fig. 4 (a) Optical micrograph of the whole micro-mixer made of two layers of TPE material bonded on a poly(cyclo)olefin polymer substrate. The device size is smaller than a centimetre square, and each valve measures $200 \times 200 \mu\text{m}$, (b) detailed view of a valve at both open and closed positions; (c) curve representing the valving cycle at 1.2 Hz, fluorescence intensity under the deflected membrane is registered, it is maximal when the valve is open and minimal when closed.

the pneumatic deflection of a $200 \times 200 \mu\text{m}$ membrane. Valve operation has been characterized with a simple observation method: a solution of fluorescein flowing under the membrane was observed with a fluorescent microscope, while the valve is actuated at different frequencies and pressures. When the valve is closed, any fluorescence solution remains and the recorded signal is minimal. Conversely, when the valve is opened, the fluorescence signal is maximal.

As can be seen in Fig. 4c, fluorescence intensity oscillations are observed for repeated cycles of opening and closing of a valve, the valve was actuated at 1.2 Hz with a control pressure of 10 PSI. Opening and closing times are 80 ms and 120 ms, respectively, corresponding to a limit frequency around 5 Hz. Otherwise, when a pressure is applied to the flowing channel, the opening time decreases and the frequency limit can be significantly improved. As depicted in Fig. 4a, devices such as micro-mixer/solution preparator were also realized following existing geometry introduced by Quake *et al.*² Briefly, when three valves are placed in series on a single channel and activated sequentially, one can pump in a peristaltic manner and thus control the movement of a small amount of fluid in a microfluidic circuitry. Moreover, when such a pump takes place on an annular channel, efficient mixing can be obtained due to streamline stacking with fluid rotation. Injection and mixing steps were thus obtained by applying the suitable pneumatic control sequence to the two peristaltic pumps of our device. Mixing of injected fluorescein solution in water was observed on the same fluorescent microscope setup. Finally, with a pumping speed around $1\text{--}5 \text{ mm s}^{-1}$, a few seconds are necessary to prepare a solution at a given concentration of fluorescein.

4. Conclusion

Undoubtedly, the expansion of microfluidics into wider applications requires plastic materials with properties beyond those of the two main avenues in use today: PDMS soft lithography and thermal embossing of rigid thermoplastics. In this report, we have demonstrated that TPE block copolymers are a third option offering convenient and enhanced microstructuration and assembling capabilities. Through the characterization of three SEBS TPE polymers and the fabrication of multilayered devices, we have demonstrated the following advantages: i) no pre-required compounding and vulcanization steps like PDMS; ii) amenability for thermoforming with high-throughput hot embossing and free pressure molding; iii) easy methods for assembling and bonding. Valves, pumps and micro-mixers were successfully fabricated and functionalities demonstrated. Therefore, we assert that TPE materials open the way for fast prototyping and potentially large fabrication of all thermo-plastic-based microfluidic systems at low cost.

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